Johns Hopkins University
Masters in Artificial Intelligence
Module 9 – CUDA Advanced Libraries

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Date: 3/20/2026

Program Goal

My goal in this project was to replace CUDA kernels with advanced libraries to simplify the code previously designed to convert RGB images to blurred grayscale images. The testing images leverage the cuRAND library to create images in the device memory directly. Next the RGB image is converted to grayscale using the library nppiRGBToGray_8u_C3C1R_Ctx and finally blurred using the library nppiFilterBoxBorder_8u_C1R_Ctx. These two functions are part of the NPP library.

Figure 1 demonstrates code compilation and execution. Five pixels are printed to show the transformation at the different steps.

```
C:\Users\Herbert\workspace\EN605.617\module9\Assignment_9c>nvcc assignment.cu -o assignment
.exe -lnppc -lnppif -lnppicc -lcurand
assignment.cu
tmpxft_00006f34_00000000-7_assignment.cudafe1.cpp

C:\Users\Herbert\workspace\EN605.617\module9\Assignment_9c>.\assignment.exe

Image size: 7680 x 4320

Test Image 1 - Sample RGB values (RGB):
RED pixels:   21 37 247 246 103
GREEN pixels: 94 90 209 230 255
BLUE pixels:  61 96 248 140 121
Gray pixels: 68 75 225 225 194
Blurred pixels: 67 117 154 185 152

Test Image 2 - Sample RGB values (RGB):
RED pixels:   245 44 173 229 229
GREEN pixels: 43 31 33 43 63
BLUE pixels:  32 13 168 200 229
Gray pixels: 102 33 90 117 132
Blurred pixels: 104 87 76 92 92

C:\Users\Herbert\workspace\EN605.617\module9\Assignment_9c>
```

Fig 1: Program compilation and execution.

My experience about the use of these CUDA libraries is that they simplify significantly the coding. They do not require the determination of grid size, number of threads, pointer creation and memory allocation in general. The abstraction provided by CUDA libraries is very useful.